R Basic Course

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Contents

About Course_R

Description and educational material for introductory course on R, descriptive statistics, and visualizations with applications in fisheries.

This course has these sections;

- 1. **useR**. Intro to R and basic features.
- 2. **Exploratory Data Analysis**. Basic concepts for data exploration.
- 3. **Descriptive Statistics**. Essential statistical calculations.
- 4. **Visualization**. Graphics and Maps using modern packages
- 5. **Efficient code**. Organization strategies.
- 6. **Automatization and sharing** Report generation and result sharing.
- 7. **Management of Bibliographic** sources and additional materials.

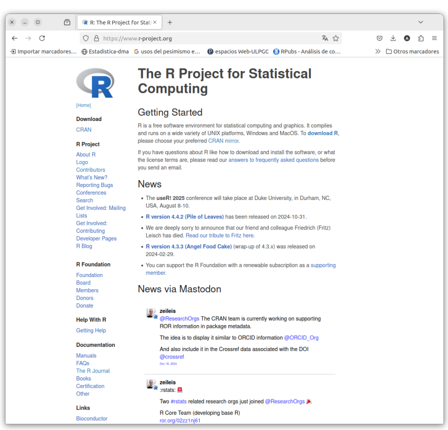
All codes and used in this course can be founded in this repository with the structure to do this Ebook Repo R Course

Chapter 1

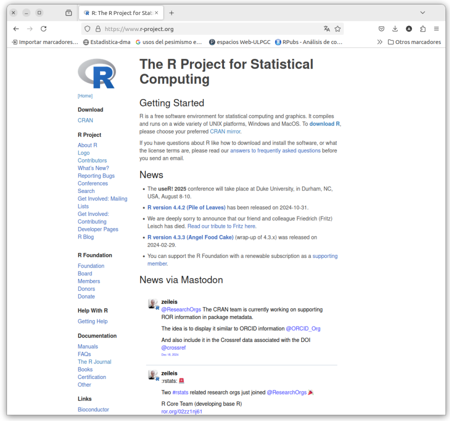
useR. Intro to R and basic features

1.1 Welcome R user

- **R** is a programming language designed for:
  - Statistical computing
  - Data analysis
  - Visualization



Download R

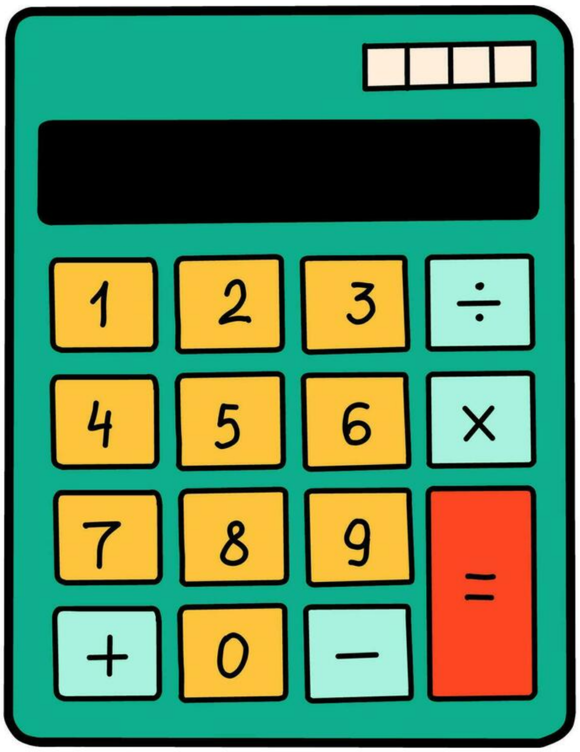


[Download Rstudio]

1.2 ¿Why learn R?

- Open source and free
- big community
- $\sim$ 13.000 packages
- Reproducible science

is just this



but…

1.3 Origin of R

Seminal paper or R! here (Fig. 1.1).

1.4 Getting Started

1. Install R from CRAN
2. Install RStudio (optional, but highly recommended)
3. Write your first R command:

```
print("Hello, world!")
```

```
## [1] "Hello, world!"
```

R: A Language for Data Analysis and Graphics

Ross IHAKA and Robert GENTLEMAN

In this article we discuss our experience designing and implementing a statistical computing language. In developing this new language, we sought to combine what we felt were useful features from two existing computer languages We feel that the new language provides advantages in the areas of portability, computational efficiency, memory management, and scoping.

Key Words: Computer language; Statistical computing.

1. INTRODUCTION

This article discusses some issues involved in the design and implementation of a computer language for statistical data analysis. Our experience with these issues occurred while developing such a language. The work has been heavily influenced by two existing languages—Becker, Chambers, and Wilks' S (1985) and Steel and Sussman's Scheme (1975). We felt that there were strong points in each of these languages and that it would be interesting to see if the strengths could be combined. The resulting language is very similar in appearance to S, but the underlying implementation and semantics are derived from Scheme. In fact, we implemented the language by first writing an interpreter for a Scheme subset and then progressively mutating it to resemble S.

We added S-like features in several stages. First, we altered the language parser so that the syntax would resemble that of S. This created a major change in the appearance of the language, but it should be emphasized that the change was entirely superficial; the underlying semantics remained those of Scheme. Next, we modified the data types of the language by removing the single scalar data type we had put into our Scheme and replacing it with the vector-based types of S. This was a much more substantive change and required major modifications to the interpreter. The final substantive change involved adding the S notion of *lazy arguments* for functions.

At this point we had enough of a framework in place to begin building a full statistical language. This process is ongoing, but we feel that we are well on the way to building a complete and useful piece of software. The development of the key portions of language

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Journal of Computational and Graphical Statistics, Volume 5, Number 3, Pages 299–314

Figure 1.1: Seminal Paper of R

1.5 Basic Data Types

- **Numeric:** Numbers (e.g., `3.14`, `42`)
- **Character:** Text strings (e.g., `"fish"`)
- **Logical:** `TRUE` or `FALSE`
- **Factor:** Categorical variables

1.6 Vectors

- R's most fundamental data structure:

```
numbers <- c(1, 2, 3, 4)
letters <- c("a", "b", "c")
```

- Operations are vectorized:

```
numbers * 2
```

```
## [1] 2 4 6 8
```

1.7 Data Frames

- Table-like structures:

```
df <- data.frame(Name = c("John", "Jane"),
    Age = c(25, 30))
```

- Inspect the structure:

```
head(df)
```

```
##   Name Age
## 1 John  25
## 2 Jane  30
```

1.8 Visualization

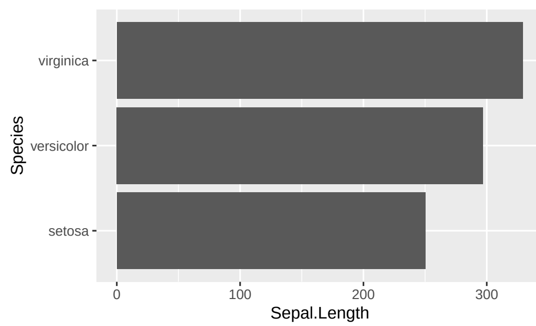
- Example base plot:

Data used

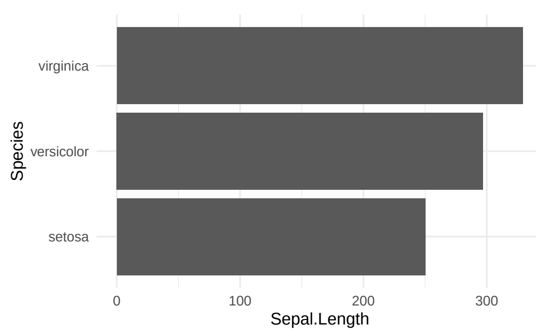
```
kbl(head(iris))
```

Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
5.1	3.5	1.4	0.2	setosa
4.9	3.0	1.4	0.2	setosa
4.7	3.2	1.3	0.2	setosa
4.6	3.1	1.5	0.2	setosa
5.0	3.6	1.4	0.2	setosa
5.4	3.9	1.7	0.4	setosa

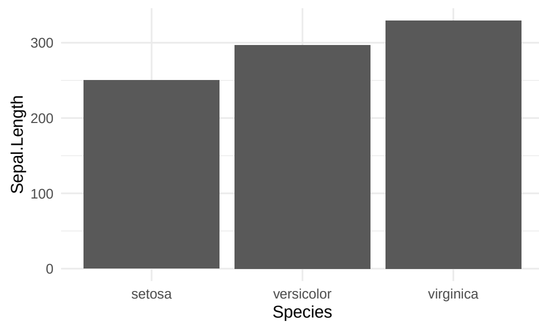
```r
plottex <- ggplot(iris, aes(x = Sepal.Length,
    y = Species)) + geom_bar(stat = "identity")
plottex
```




```r
plottex <- ggplot(iris, aes(x = Sepal.Length,
    y = Species)) + geom_bar(stat = "identity") +
    theme_minimal()
plottex
```




```r
plottex <- ggplot(iris, aes(x = Sepal.Length,
    y = Species)) + geom_bar(stat = "identity") +
    theme_minimal() + coord_flip()
plottex
```



1.9 Functions

- Encapsulate reusable logic:

```
square <- function(x) {
    return(x^2)
}
square(4)  # Output: 16
```

```
## [1] 16
```

1.10 Statistical Analysis

- Linear regression example:

```
fit <- lm(mpg ~ wt, data = mtcars)
```

```
summary(fit)
```

```
##
## Call:
## lm(formula = mpg ~ wt, data = mtcars)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.5432 -2.3647 -0.1252  1.4096  6.8727
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  37.2851     1.8776  19.858  < 2e-16 ***
## wt           -5.3445     0.5591  -9.559 1.29e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.046 on 30 degrees of freedom
```

```
## Multiple R-squared:  0.7528, Adjusted R-squared:  0.7446
## F-statistic: 91.38 on 1 and 30 DF,  p-value: 1.294e-10
```

1.11 Conclusion

- R is powerful for:
  - Data analysis
  - Visualization
  - Statistics

1.12 Sources

Source in this link

An introduction to R

Chapter 2

Exploratory Data Analysis

Exploratory Data Analysis (EDA) is a crucial first step in analyzing data. It provides a summary of the main characteristics of datasets and helps identify patterns, outliers, and relationships.

libraries

```r
library(tidyverse)
```

2.1 Summary Statistics in R

One of the simplest ways to start analyzing data is to generate summary statistics.

```r
# Simple summary of built-in dataset
summary(mtcars)
```

```
##       mpg             cyl             disp             hp       
##  Min.   :10.40   Min.   :4.000   Min.   : 71.1   Min.   : 52.0  
##  1st Qu.:15.43   1st Qu.:4.000   1st Qu.:120.8   1st Qu.: 96.5  
##  Median :19.20   Median :6.000   Median :196.3   Median :123.0  
##  Mean   :20.09   Mean   :6.188   Mean   :230.7   Mean   :146.7  
##  3rd Qu.:22.80   3rd Qu.:8.000   3rd Qu.:326.0   3rd Qu.:180.0  
##  Max.   :33.90   Max.   :8.000   Max.   :472.0   Max.   :335.0  
##       drat             wt             qsec             vs        
##  Min.   :2.760   Min.   :1.513   Min.   :14.50   Min.   :0.0000  
##  1st Qu.:3.080   1st Qu.:2.581   1st Qu.:16.89   1st Qu.:0.0000  
##  Median :3.695   Median :3.325   Median :17.71   Median :0.0000  
##  Mean   :3.597   Mean   :3.217   Mean   :17.85   Mean   :0.4375  
##  3rd Qu.:3.920   3rd Qu.:3.610   3rd Qu.:18.90   3rd Qu.:1.0000  
##  Max.   :4.930   Max.   :5.424   Max.   :22.90   Max.   :1.0000  
```

```
##        am              gear            carb      
##  Min.   :0.0000   Min.   :3.000   Min.   :1.000  
##  1st Qu.:0.0000   1st Qu.:3.000   1st Qu.:2.000  
##  Median :0.0000   Median :4.000   Median :2.000  
##  Mean   :0.4062   Mean   :3.688   Mean   :2.812  
##  3rd Qu.:1.0000   3rd Qu.:4.000   3rd Qu.:4.000  
##  Max.   :1.0000   Max.   :5.000   Max.   :8.000  
```

2.2 Data Handling

2.2.1 Loading and Inspecting Data

EDA often begins by loading the dataset and inspecting its structure.

```
# Load CSV data
url <- "https://people.sc.fsu.edu/~jburkardt/data/csv/hw_200.csv"
data <- read.csv(url)

# View first rows of the data
head(data)
```

```
##   Index
## 1     2
## 2     3
## 3     4
## 4     5
## 5     6
## 6     7
##   Height.Inches...Weight.Pounds..1..65.78..112.99.2..71.52..136.49.3..69.40..153.03
## 1
## 2
## 3
## 4
## 5
## 6
##   Weight.Pounds.
## 1         136.49
## 2         153.03
## 3         142.34
## 4         144.30
## 5         123.30
## 6         141.49
```

2.3 Data Manipulation

2.3.1 Subsetting Data

You can select specific columns or rows based on logical operations.

```
# Select rows where mpg is greater than
# 20
subset_mpg <- mtcars[mtcars$mpg > 20, ]
head(subset_mpg)
```

```
##                 mpg cyl  disp  hp drat    wt  qsec vs am gear carb
## Mazda RX4      21.0   6 160.0 110 3.90 2.620 16.46  0  1    4    4
## Mazda RX4 Wag  21.0   6 160.0 110 3.90 2.875 17.02  0  1    4    4
## Datsun 710     22.8   4 108.0  93 3.85 2.320 18.61  1  1    4    1
## Hornet 4 Drive 21.4   6 258.0 110 3.08 3.215 19.44  1  0    3    1
## Merc 240D      24.4   4 146.7  62 3.69 3.190 20.00  1  0    4    2
## Merc 230       22.8   4 140.8  95 3.92 3.150 22.90  1  0    4    2
```

2.3.2 Using dplyr for Data Manipulation

The `dplyr` package makes data manipulation efficient and readable.

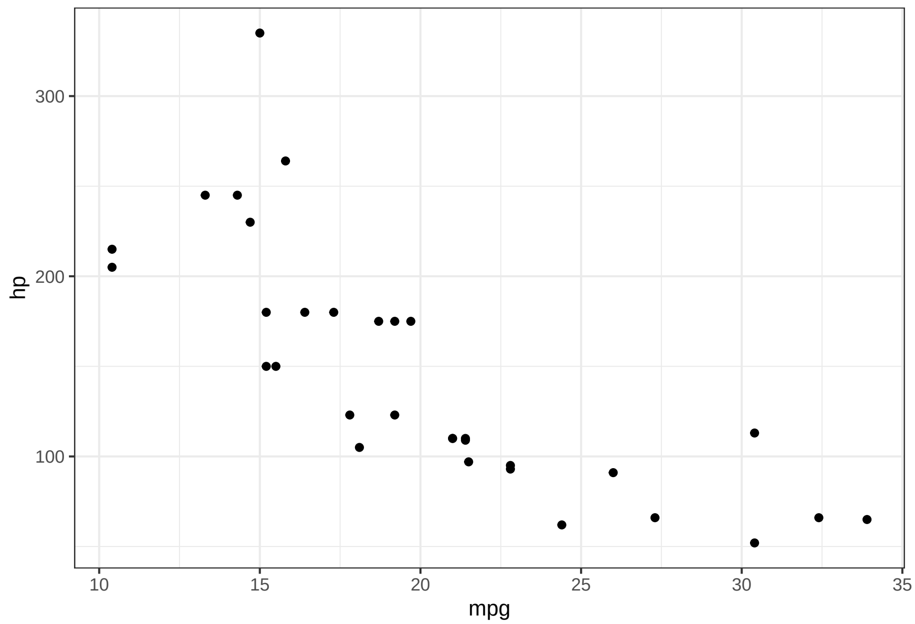
```
# Load the dplyr package
library(dplyr)

# Filter and summarize data
mtcars %>%
    filter(mpg > 20) %>%
    summarise(avg_hp = mean(hp), count = n())
```

```
##   avg_hp count
## 1   88.5    14
```

2.4 Visualization with Base R

Visualizations are essential for gaining insights quickly.

```
# Simple scatter plot
carsplot <- ggplot() + geom_point(data = mtcars,
    aes(mpg, hp)) + theme_bw()
carsplot
```



2.5 Final Thoughts

- EDA is critical for understanding your data.
- Data handling ensures clean datasets.
- Data manipulation prepares data for in-depth analysis.
- Visualization helps reveal insights quickly.

2.6 References

Basic R

Data Explorer

EMSE 4572 CleanData

Chapter 3

Descriptive Statistics.

Descriptive statistics summarize and organize data. They help us understand patterns and key characteristics in a dataset.

We'll explore basic concepts using the `penguins` dataset from the `palmerpenguins` package.

3.1 The `penguins` Dataset

```r
library(tidyverse)
library(palmerpenguins)
library(kableExtra)
data("penguins")
kbl(head(penguins, n = 4))
```

species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	sex	year
Adelie	Torgersen	39.1	18.7	181	3750	male	2007
Adelie	Torgersen	39.5	17.4	186	3800	female	2007
Adelie	Torgersen	40.3	18.0	195	3250	female	2007
Adelie	Torgersen	NA	NA	NA	NA	NA	2007

3.2 Basic Data Exploration

```r
# Check dimensions and structure
dim(penguins)
```

```
## [1] 344   8
```

```r
str(penguins)
```

```
## tibble [344 x 8] (S3: tbl_df/tbl/data.frame)
##  $ species          : Factor w/ 3 levels "Adelie","Chinstrap",..: 1 1 1 1 1 1 1 1 1
##  $ island           : Factor w/ 3 levels "Biscoe","Dream",..: 3 3 3 3 3 3 3 3 3 3 .
##  $ bill_length_mm   : num [1:344] 39.1 39.5 40.3 NA 36.7 39.3 38.9 39.2 34.1 42 ...
##  $ bill_depth_mm    : num [1:344] 18.7 17.4 18 NA 19.3 20.6 17.8 19.6 18.1 20.2 ...
##  $ flipper_length_mm: int [1:344] 181 186 195 NA 193 190 181 195 193 190 ...
##  $ body_mass_g      : int [1:344] 3750 3800 3250 NA 3450 3650 3625 4675 3475 4250 .
##  $ sex              : Factor w/ 2 levels "female","male": 2 1 1 NA 1 2 1 2 NA NA ..
##  $ year             : int [1:344] 2007 2007 2007 2007 2007 2007 2007 2007 2007 2007
```

3.3 Summary Statistics

```r
# Quick statistical overview
summary(penguins)
```

```
##       species          island    bill_length_mm  bill_depth_mm  
##  Adelie   :152   Biscoe   :168   Min.   :32.10   Min.   :13.10  
##  Chinstrap: 68   Dream    :124   1st Qu.:39.23   1st Qu.:15.60  
##  Gentoo   :124   Torgersen: 52   Median :44.45   Median :17.30  
##                                  Mean   :43.92   Mean   :17.15  
##                                  3rd Qu.:48.50   3rd Qu.:18.70  
##                                  Max.   :59.60   Max.   :21.50  
##                                  NA's   :2       NA's   :2      
##  flipper_length_mm  body_mass_g       sex           year     
##  Min.   :172.0     Min.   :2700   female:165   Min.   :2007  
##  1st Qu.:190.0     1st Qu.:3550   male  :168   1st Qu.:2007  
##  Median :197.0     Median :4050   NA's  : 11   Median :2008  
##  Mean   :200.9     Mean   :4202                Mean   :2008  
##  3rd Qu.:213.0     3rd Qu.:4750                3rd Qu.:2009  
##  Max.   :231.0     Max.   :6300                Max.   :2009  
##  NA's   :2         NA's   :2
```

3.4 Central Tendency

- **Mean:** average value
- **Median:** middle value
- **Mode:** most frequent value

```r
mean(penguins$bill_length_mm, na.rm = TRUE)
```

```
## [1] 43.92193
```

```r
median(penguins$bill_length_mm, na.rm = TRUE)
```

```
## [1] 44.45
```

3.5 Variability

- **Range:** difference between max and min
- **Variance:** spread of the data
- **Standard deviation:** square root of variance

```r
range(penguins$bill_length_mm, na.rm = TRUE)
```

```
## [1] 32.1 59.6
```

```r
var(penguins$bill_length_mm, na.rm = TRUE)
```

```
## [1] 29.80705
```

```r
sd(penguins$bill_length_mm, na.rm = TRUE)
```

```
## [1] 5.459584
```

3.6 Grouped Summaries

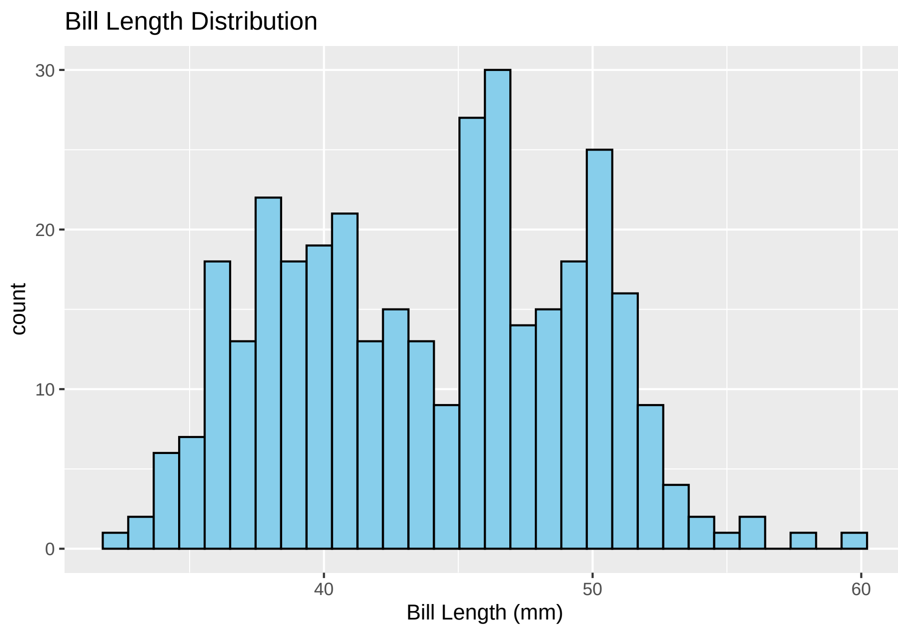
```r
# Grouped mean by species
library(dplyr)
penguins %>%
    group_by(species) %>%
    summarise(mean_bill_length = mean(bill_length_mm,
        na.rm = TRUE))
```

```
## # A tibble: 3 x 2
##   species   mean_bill_length
##   <fct>                <dbl>
## 1 Adelie                38.8
## 2 Chinstrap             48.8
## 3 Gentoo                47.5
```

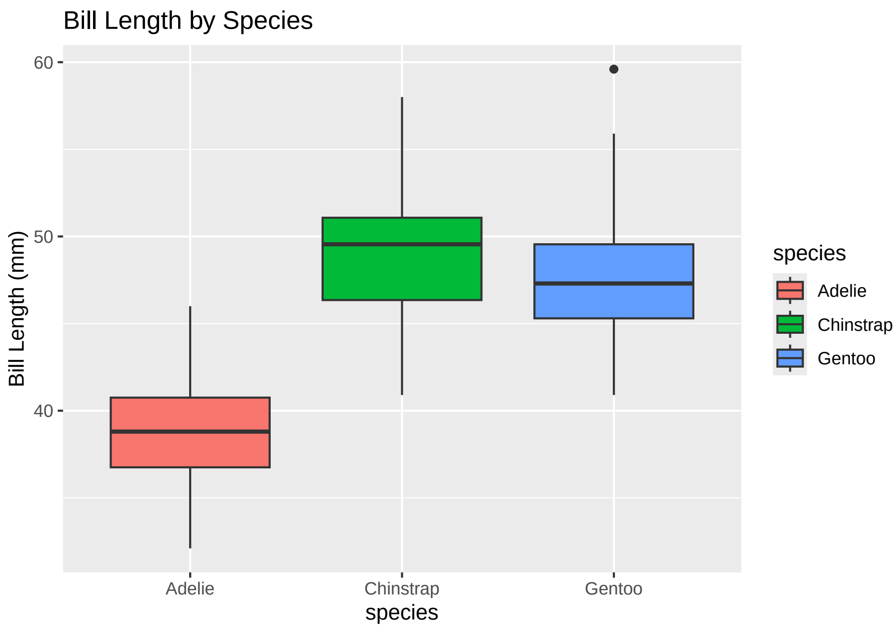
3.7 Visualization: Histograms

```r
library(ggplot2)
ggplot(penguins, aes(x = bill_length_mm)) +
    geom_histogram(bins = 30, fill = "skyblue",
```

```
        color = "black") + labs(title = "Bill Length Distribution",
    x = "Bill Length (mm)")
```




```
ggplot(penguins, aes(x = species, y = bill_length_mm,
    fill = species)) + geom_boxplot() + labs(title = "Bill Length by Species",
    y = "Bill Length (mm)")
```



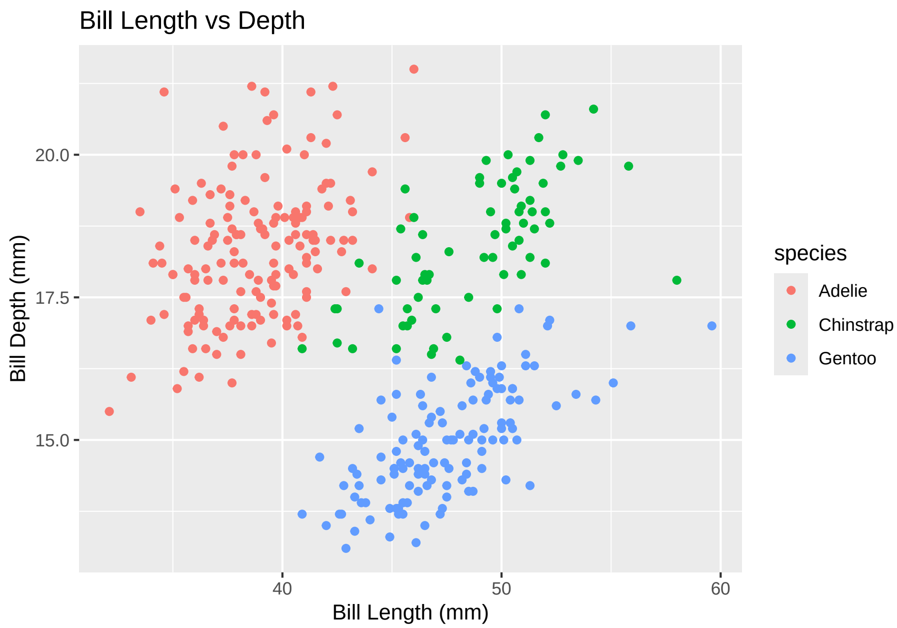
3.9 Correlation

```r
cor(penguins$bill_length_mm, penguins$bill_depth_mm,
    use = "complete.obs")
```

```
## [1] -0.2350529
```

3.10 Scatter Plot for Relationships

```r
ggplot(penguins, aes(x = bill_length_mm,
    y = bill_depth_mm, color = species)) +
    geom_point() + labs(title = "Bill Length vs Depth",
    x = "Bill Length (mm)", y = "Bill Depth (mm)")
```



3.11 Conclusion

- Descriptive statistics are fundamental for understanding data.
- R provides powerful tools to calculate, visualize, and interpret data patterns.

Questions?

Sources in this link

here

Data Types & Probability Distributions

Chapter 4

Visualization.

Graphics and Maps using modern packages

4.1 Graphics

Footnotes are put inside the square brackets after a caret `^[]`. Like this one [1].

4.2 Simple maps

Reference items in your bibliography file(s) using `@key`.

For example, we are using the **bookdown** package [Xie, 2024] (check out the last code chunk in index.Rmd to see how this citation key was added) in this sample book, which was built on top of R Markdown and **knitr** [Xie, 2015] (this citation was added manually in an external file book.bib). Note that the `.bib` files need to be listed in the index.Rmd with the YAML `bibliography` key.

The RStudio Visual Markdown Editor can also make it easier to insert citations: https://rstudio.github.io/visual-markdown-editing/#/citations

[1] This is a footnote.

Chapter 5

Efficient Code

5.1 Equations

Here is an equation.

$$f\left(k\right) = \binom{n}{k} p^k \left(1-p\right)^{n-k} \tag{5.1}$$

You may refer to using `\@ref(eq:binom)`, like see Equation (5.1).

Read more here https://bookdown.org/yihui/bookdown/markdown-extensions-by-bookdown.html.

5.2 Callout blocks

The R Markdown Cookbook provides more help on how to use custom blocks to design your own callouts: https://bookdown.org/yihui/rmarkdown-cookbook/custom-blocks.html

Chapter 6

Reporting and Sharing

6.1 Publishing

HTML books can be published online, see: https://bookdown.org/yihui/bookdown/publishing.html

6.2 404 pages

By default, users will be directed to a 404 page if they try to access a webpage that cannot be found. If you'd like to customize your 404 page instead of using the default, you may add either a `_404.Rmd` or `_404.md` file to your project root and use code and/or Markdown syntax.

6.3 Metadata for sharing

Bookdown HTML books will provide HTML metadata for social sharing on platforms like Twitter, Facebook, and LinkedIn, using information you provide in the `index.Rmd` YAML. To setup, set the `url` for your book and the path to your `cover-image` file. Your book's `title` and `description` are also used.

This `gitbook` uses the same social sharing data across all chapters in your book-all links shared will look the same.

Specify your book's source repository on GitHub using the `edit` key under the configuration options in the `_output.yml` file, which allows users to suggest an edit by linking to a chapter's source file.

Read more about the features of this output format here:

https://pkgs.rstudio.com/bookdown/reference/gitbook.html

Or use:

```
`?`(bookdown::gitbook)
```

Bibliography

Yihui Xie. *Dynamic Documents with R and knitr*. Chapman and Hall/CRC, Boca Raton, Florida, 2nd edition, 2015. URL http://yihui.org/knitr/. ISBN 978-1498716963.

Yihui Xie. *bookdown: Authoring Books and Technical Documents with R Markdown*, 2024. URL https://github.com/rstudio/bookdown. R package version 0.40.